

EXPERIMENT – 14

Study and Demonstration of Brute Force Password Attack in a Controlled Kali Linux Environment

Aim

To study and demonstrate the concept of brute force password attacks in a controlled Kali Linux environment to understand password vulnerabilities and the importance of strong authentication mechanisms.

Software Requirements

- **Host Operating System:** Windows / Linux / macOS
- **Virtualization Software:** Oracle VirtualBox
- **Guest Operating System:** Kali Linux
- **Study Tools:** Open-source password auditing utilities (educational use only)
- **System Requirements:**
 - RAM: Minimum 4 GB (8 GB recommended)
 - Disk Space: Minimum 30 GB
 - Processor: Intel/AMD with Virtualization support

Introduction

A brute force password attack is a method where all possible password combinations are systematically tried until the correct one is found.

Such attacks exploit weak passwords, poor authentication policies, and lack of account lockout mechanisms.

Studying brute force techniques in a controlled Kali Linux environment helps learners understand real-world risks and reinforces best practices for password security and system hardening.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start Kali Linux.
3. Log in using Kali credentials.

Step 2: Set Up a Controlled Test Environment

1. Use a **locally created test account** or **dummy authentication service**.
2. Ensure no real systems or credentials are involved.
3. Confirm the environment is isolated from external networks.

Step 3: Study Brute Force Attack Workflow

1. Understand how login mechanisms validate credentials.
2. Analyze how attackers automate repeated login attempts.
3. Observe how password complexity affects attack feasibility.

Step 4: Educational Demonstration (Simulation Only)

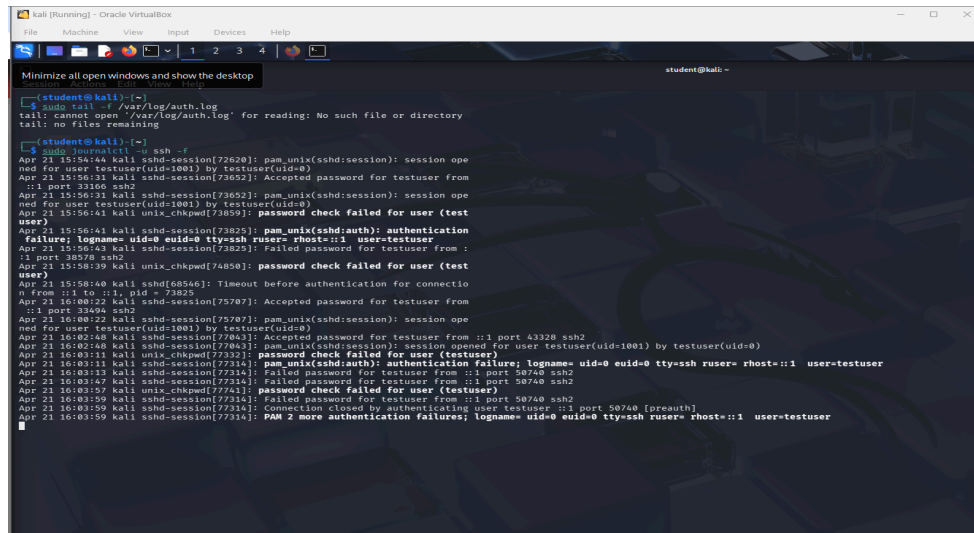
1. Simulate repeated authentication attempts using dummy data.
2. Observe system responses such as delays or access denial.
3. Record observations without executing real attacks.

Step 5: Analyze Security Weaknesses

- Weak or short passwords
- No rate limiting
- Missing account lockout policies
- Lack of multi-factor authentication

Step 6: Study Prevention and Mitigation Techniques

- Strong password policies
- Account lockout and rate limiting
- CAPTCHA mechanisms
- Multi-factor authentication (MFA)
- Monitoring and alerting



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(student@kali) ~$ sudo tail -f /var/log/auth.log
tail: cannot open '/var/log/auth.log' for reading: No such file or directory
tail: no files remaining

(student@kali) ~$ sudo journalctl --no-pager --output=short -u ssh -f
Apr 21 15:54:44 kali sshd-session[72620]: pam_unix(sshd:session): session opened for user testuser(uid=1001) by testuser(uid=0)
Apr 21 15:56:31 kali sshd-session[73652]: Accepted password for testuser from ::1 port 33166 ssh2
Apr 21 15:56:31 kali sshd-session[73652]: pam_unix(sshd:session): session opened for user testuser(uid=1001) by testuser(uid=0)
Apr 21 15:56:41 kali unix_chkpwd[73859]: password check failed for user (testuser)
Apr 21 15:56:41 kali sshd-session[73825]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=::1 user=testuser
Apr 21 15:56:43 kali sshd-session[73825]: Failed password for testuser from ::1 port 38578 ssh2
Apr 21 15:58:39 kali unix_chkpwd[74858]: password check failed for user (testuser)
Apr 21 15:58:40 kali sshd[68546]: Timeout before authentication for connection from ::1 to ::1, pid = 73825
Apr 21 16:00:22 kali sshd-session[75787]: Accepted password for testuser from ::1 port 33494 ssh2
Apr 21 16:00:22 kali sshd-session[75787]: pam_unix(sshd:session): session opened for user testuser(uid=1001) by testuser(uid=0)
Apr 21 16:02:48 kali sshd-session[77043]: Accepted password for testuser from ::1 port 43328 ssh2
Apr 21 16:02:48 kali sshd-session[77043]: pam_unix(sshd:session): session opened for user testuser(uid=1001) by testuser(uid=0)
Apr 21 16:03:11 kali unix_chkpwd[77332]: password check failed for user (testuser)
Apr 21 16:03:11 kali sshd-session[77314]: pam_unix(sshd:auth): authentication failure; logname=uid=0 euid=0 tty=ssh ruser= rhost=::1 user=testuser
Apr 21 16:03:13 kali sshd-session[77314]: Failed password for testuser from ::1 port 50740 ssh2
Apr 21 16:03:47 kali sshd-session[77314]: Failed password for testuser from ::1 port 50740 ssh2
Apr 21 16:03:57 kali unix_chkpwd[77741]: password check failed for user (testuser)
Apr 21 16:03:59 kali sshd-session[77314]: Failed password for testuser from ::1 port 50740 ssh2
Apr 21 16:03:59 kali sshd-session[77314]: Connection closed by authenticating user testuser ::1 port 50740 [preauth]
Apr 21 16:03:59 kali sshd-session[77314]: PAM 2 more authentication failures; logname=uid=0 euid=0 tty=ssh ruser= rhost=::1 user=testuser
  
```

Result

The study and demonstration of brute force password attacks were successfully completed in a controlled Kali Linux environment.

The experiment emphasized the importance of strong passwords and robust authentication mechanisms to prevent unauthorized access.